

ZHONGKAI YU

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Education

2024 – Now **Ph.D in Computer Science and Engineering**, University of California, San Diego (UCSD), USA

Advisor: Prof. [Yufei Ding](#)

2021 - 2024 **M.E. in Computer Technology**, Chinese Academy of Sciences (ICT, CAS), China

Advisor: Prof. [Yunji Chen](#)

2017 – 2021 **B.S. in Physics** (Zhiyuan Honors Program), Shanghai Jiao Tong University (SJTU), China

2019 – 2019 **Exchange Student** in Physics / EECS, Massachusetts Institute of Technology (MIT), USA

Research Interest

My research centers on the co-evolution of LLM and hardware systems: I explore novel architectures and systems to accelerate LLM workloads, while harnessing LLMs to advance chip design. My work lies at the intersection of Computer Architecture, ML Systems, and AI4Chip.

Publications

[ISCA' 26](#)

Zhongkai Yu, Yue Guan, Zihao Yu, Chenyang Zhou, Zhengding Hu, Shuyi Pei, Yangwook Kang, Yufei Ding, Po-An Tsai. "Patterns behind Chaos: Forecasting Data Movement for Efficient Large-Scale MoE LLM Inference" 🏆 **Best Paper Award** (2/845)

[MICRO' 24](#)

Zhongkai Yu*, Shengwen Liang*, Tianyun Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, Jie Zhang, Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu, Qi Guo, Tianshi Chen. "Cambricon-LLM: A Chiplet-Based Hybrid Architecture for On-Device Inference of 70B LLM"

[ArXiv](#)

[Submitted]

Zhongkai Yu, Haotian Ye, Chenyang Zhou, Ohm Rishabh Venkatachalam, Zaifeng Pan, Zhengding Hu, Junsung Kim, Won Woo Ro, Po-An Tsai, Shuyi Pei, Yangwook Kang, Yufei Ding. "AMMA: A Multi-Chiplet Memory-Centric Architecture for Low-Latency 1M Context Attention Serving"

[Submitted]

Zhongkai Yu, Ohm Rishabh Venkatachalam, Zheng Wang, Yikai Li, Yichen Lin, Zihao Yu, Yuke Wang, Liu Liu, Xulong Tang, Shuyi Pei, Yangwook Kang, Yufei Ding. "CLoRA: A CXL-Based System for Cost-Efficient Multi-LoRA Serving"

[ArXiv](#)

[Submitted]

Zhongkai Yu*, Chenyang Zhou*, Yichen Lin, Hejia Zhang, Haotian Ye, Junxia Cui, Zaifeng Pan, Jishen Zhao, Yufei Ding. "ChipBench: A Next-Step Benchmark for Evaluating LLM Performance in AI-Aided Chip Design"

[ArXiv](#)

[Submitted]

Zhongkai Yu*, Yichen Lin*, Chenyang Zhou, Yuwei Zhang, Kun Zhou, Junxia Cui, Haotian Ye, Zhengding Hu, Ruiyi Wang, Yujie Zhao, Hejia Zhang, Jingbo Shang, Jishen Zhao, Yufei Ding. "ChipMATE: Multi-Agent Training via Reinforcement Learning for Enhanced RTL Generation"

TACO [Submitted]	Zheng Wang*, Zhongkai Yu* , Kaijian Wang, Yichen Lin, Yikai Li, Liu Liu, Xulong Tang, Yuke Wang, Yangwook Kang, Yufei Ding. "ReDM: Efficient Training System for Large-Scale Recommendation Models on Disaggregated Memory"
OSDI' 25	Yue Guan, Yuanwei Fang, Keren Zhou, Corbin Rebeck, Manman Ren, Zhongkai Yu , Yufei Ding, Adnan Aziz. "KPerfIR: Towards an Open and Compiler-centric Ecosystem for GPU Kernel Performance Tooling on Modern AI Workloads"
DAC' 22	Zhuoran Song, Zhongkai Yu , Naifeng Jing, Xiaoyao Liang. "E2SR: an end-to-end video CODEC assisted system for super resolution acceleration", Proceedings of the 59 th ACM/IEEE Design Automation Conference, 2022
TCAD	Tianyun Ma, Yuanbo Wen, Xinkai Song, Pengwei Jin, Di Huang, Husheng Han, Ziyuan Nan, Zhongkai Yu , Shaohui Peng, Yongwei Zhao, Huaping Chen, Zidong Du, Xing Hu and Qi Guo. Harmonia: A Unified Architecture for Efficient Deep Symbolic Regression.

Internship

NVIDIA Research

Jun 2026 – Sep 2026

Research Intern, Architecture Group, Manager: [Po-An Tsai](#), [Jenny Huang](#), and [Christos Kozyrakis](#)
Santa Clara, US

- Waferscale GPU architecture and system design for LLM

Samsung Semiconductor

Jun 2025 – Sep 2025

Research Intern, AGI Lab, Manager: [Shuyi Pei](#) and [Yangwook Kang](#)

San Jose, US

- MoE serving on multi-chiplet HBM-NMP architectures

Cambricon Technologies

Mar 2022 – Mar 2024

Digital IC Design Intern, Architecture Group, Manager: [Dong Han](#) and [Shaoli Liu](#)

Beijing, China

- Digital Front-end Design of a Commercial Cloud AI Chip Using Advanced Process
- Max-Power Test Cases for a Commercial AI Chip Using High-level Programming Language

Selected Projects

Patterns behind Chaos: Forecasting Data Movement for Efficient Large-Scale MoE LLM Inference

- Conducted a comprehensive data-movement-centric profiling of top tier MoE models (200B - 671B)
- Made systematic analysis and distilled six insights applicable for varied serving system design
- Proposed a future GPU architecture design to verify the insights

CLoRA: Cost-effective System for Multi-LoRA Model Inference using CXL

- Proposed a CXL-based near data processing architecture to support collaboration with GPU.
- Designed multiple collaborative approaches to utilize both GPU and NDP for multi-LoRA serving
- Built strategy selection and memory management system for optimal hardware utilization .

Cambricon-LLM: Accelerator for On-device Inference of 70B LLM

- Designed an innovative architecture to enable the inference of 70B-LLM on small edge devices.

- Reached a speed of 3.44 tokens for 70B LLM, and 22x to 45x speedup over existing technologies.

Collaborative Projects

KPerfIR: A Compiler-centric Profiler for GPU Kernel Performance Tooling

- Participated in building a compiler-centric profiler and tested it on NV/AMD GPUs.
- Used the profiler to improve naïve Triton FA-3 kernel

ReDM: CXL-based Memory Disaggregated System for Large-scale Recommendation Models

- Modified the original CXL memory extender architecture to support near data processing.
- Designed an access-aware strategy for offloading data to CXL memory and achieved a 2.37x speedup over traditional workload.

E2SR: An End-to-End CODEC Assisted System for Super Resolution (SR) Acceleration

- Enhanced SR algorithm acceleration on edge devices utilizing the computing power on cloud server.
- Developed E2SR/E2SR+ architecture on edge device to support end-to-end system integration.
- Proposed an adaptive framework responsive to varied environmental conditions.

Skills

Languages: Python, C, MATLAB, Verilog, CUDA, BANGC

Tools: Linux, (g)vim, Latex, VCS, Origin, Mathematica